

StarAI - Agentic Customer Support System Architecture

1. Problem Statement

PointStar provides AI Transformation, Cloud Computing, and IT Consulting services. The following challenges should be addressed:

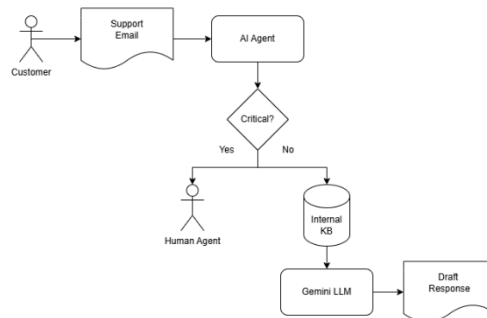
- 1) Customer inquiries may relate to service implementation, system configuration, or technical issues.
- 2) Emails should be accurately categorized and prioritized based on their urgency.
- 3) Responses should be generated using reliable information from internal documentation.
- 4) Manual processing may lead to delayed responses, incorrect classification, and improper handling of critical issues.

2. Proposed Solution

To address these challenges, the proposed system is designed to:

- 1) Classify incoming emails into appropriate categories.
- 2) Detect and flag critical issues before response drafting.
- 3) Retrieve relevant information from the internal knowledge base.
- 4) Generate accurate and consistent draft responses.
- 5) Escalate critical cases (e.g., service outages, data loss, or security breaches) to human support agents.

3. System Architecture



4. Design Considerations

The system is designed with the following considerations:

- 1) Accurately classify customer emails based on their categories.
- 2) Generate responses using information retrieved from the internal knowledge base.
- 3) Detect and escalate critical issues (e.g., data loss, service outages, and security breaches) to human support agents.
- 4) Adopt a modular architecture to simplify system maintenance and future enhancements.

5. Limitation & Mitigation

Potential failure points and mitigation strategies include:

- 1) Missing Information: If the required information is unavailable in the internal knowledge base, the request is escalated to a human support agent.
- 2) Ambiguous or Incomplete Emails: Emails with insufficient information may affect classification accuracy and require additional review.
- 3) Incorrect Response Generation: Responses are generated only from information available in the internal knowledge base.
- 4) Critical Issues: Emails related to data loss, service outages, security breaches, or repeated support requests are automatically routed to a human support agent.